

Exercise sheet 1

Probability and Statistics, MTH102

1. Show from the axioms that $P(\emptyset) = 0$.
2. If a coin is tossed 3 times, compute and observe the probability of getting
 - (a) 0 heads
 - (b) 1 head
 - (c) 2 heads
 - (d) 3 heads
 - (e) What do you think the sum of the probabilities that you computed above should be?
3. If I roll a pair of dice, and add the sum of the numbers on the dice, which number has the highest probability of occurring and why?
4. If you feel that the probability that it will rain tomorrow is 0.3 and you feel that probability that I will be absent tomorrow is 0.4. Let us assume that you feel that the probability that it rains tomorrow and that I will be absent is 0.2. What is the probability that...
 - (a) I will be present tomorrow and it will rain tomorrow?
 - (b) I will be absent tomorrow and that it will rain tomorrow?
 - (c) I will be present tomorrow or that it will not rain tomorrow?
5. Consider a coin tossed n times.
 - (a) What is the probability of getting k tails?
 - (b) Use this to show that ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$
 - (c) If n is even, what is the probability of getting heads in exactly half the number tosses?
 - (d) If n is odd, use the previous part to show that ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_{\lfloor n/2 \rfloor} = 2^{n-1}$
6. Recall that we defined the “conditional probability” $P(A|B)$ which denotes “probability of A given that we know that B has occurred” by $P(A|B) = \frac{P(A \cap B)}{P(B)}$. If two events are independent, it means that the probability of A does not depend on B, i.e. $P(A|B) = P(A)$.
 - (a) Show that if the A and B are independent then $P(A \cap B) = P(A)P(B)$
 - (b) Show that if $P(A|B) = P(A)$, then $P(B|A) = P(B)$.

- (c) If a coin is tossed twice, let A_1 denote the event of getting a head in the first toss, and A_2 denote the event of getting a head in the second toss. We may assume that they are independent events. Then use the above discussions to show that the probability of getting a head in both tosses is $1/4$.
- (d) If a coin is tossed n times, use the above observations to compute the probability of getting all heads.
- (e) Derive a formula for $P(A|B)$ in terms of $P(B|A)$, $P(A)$, and $P(B)$. (*Hint:* eliminate the $P(A \cap B)$ in the definitions of the conditional probabilities).

Exercise sheet 2

Probability and Statistics, MTH102

1. Show that if the probability of $P(E|F) = P(E|F^c)$ then E is independent of F . In other words, if the probability of E remains unchanged whether F occurs or does not occur, then E must be independent of F .
2. Show that if $P(E|F^c) = 0$, then $P(F|E) = 1$. Can you give a real life interpretation of this?
3. Show that $P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$
4. Show that $P(E_1 \cap E_2 \cap E_3 \cap E_4) = P(E_1)P(E_2|E_1)P(E_3|E_1 \cap E_2)P(E_4|E_1 \cap E_2 \cap E_3)$
5. If a deck of playing cards is divided into 4 piles of 13 cards each then what is the probability that each pile has at least one ace.

Exercise sheet 3

Probability and Statistics, MTH102

1. If a pair of dice is rolled and we know that the numbers on it are different, what is the probability that one of them is a 4?
2. Suppose 90% of all the students who attend the lectures regularly pass a certain course. While, only 30% of students who do not attend the lectures regularly pass the course. If I know that a student passed the course, what is the probability that that student attended regularly? Assume that half the students attended the course regularly.
3. Prove that if E and F are independent events, then so is E and F^c .
4. If three cards are in a box so that one card has both sides black, another has both sides white, and the remaining card has one side black and one side white. If a card is picked at random and is found to have one side white, what is the probability that the other side is black?
5. A scientist has four competing hypothesis, say H_1 , H_2 , H_3 , and H_4 for a particular phenomenon. Initially, the scientist assumes that H_1 is very likely and believes the probability to be around 0.7. However, the scientist believes that the other 3 hypotheses are equally likely. To test this the scientist performs an experiment and obtains some outcome E . Unfortunately, the experiment is not foolproof. If H_1 were true, then the probability of the experimental outcome being E is 0.1. If H_2 were true, then E will still occur with probability 0.8. If H_3 were true, then the probability of the experimental outcome being E is 0.2, and if H_4 were true, then the probability of the experimental outcome being E is 0.1. How should the scientist update the probabilities of H_1 , H_2 , H_3 , and H_4 , now that the experiment has been performed and resulted in outcome E . Which hypothesis does the scientist now believe is most probable? Which hypothesis does the scientist now believe is most probable?

Exercise sheet 4

Probability and Statistics, MTH102

1. If a pair of dice are rolled. Let X denote the sum of the values on the dice. Compute $E[X]$.
2. A pair of dice are rolled. Every time at least one of the dice shows up a 6, a person gets 10 points, and every time at least one of the dice shows up a 1, a person 2 points are deducted. For anything else, there is no change in the points. Represent this as a random variable X and compute $E[X]$.
3. Let X denote the the difference between the number of heads and tails when tossing a coin n times. What are the possible values that the random variable X can take? If $n = 4$, what is the probability mass function?
4. If X is a random variable that can take the values $-1, 0$, or 1 . Let $P(X = -1) = 1/7$, $P(X = 0) = 2/7$, and $P(X = 1) = 4/7$. Let Y denote the random variable so that $Y = X^2$. Compute $E[Y]$.
5. A biased coin with probability of head appearing being p is tossed n times. Let X denote the number of times a head appeared. Assume $E[X^2] = \sum x_i^2 p(x_i)$ and compute $E[X^2]$

Exercise sheet 5

Probability and Statistics, MTH102

1. For some random variable X , and some real numbers a and b , show that $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
2. Consider a random variable for which $E[X] = 1$ and $\text{Var}(X) = 5$. Then compute $E[(2 + X)^2]$ and $\text{Var}(4 + 3X)$.
3. Let X denote the random variable which is the number of times a coin lands on heads if it is tossed n times. Show that $P\{X = k + 1\} = \frac{p}{1-p} \frac{n-k}{k+1} P\{X = k\}$
4. Use the above to compute the approximate probability of getting k heads, for each k , for an unbiased coin which is tossed 5 times.
5. A multiple choice exam has 20 questions and each question has exactly 4 choices out of which only one is correct.
 - (a) If a student guesses all the answers, what is the probability that the person will get all the answers correct?
 - (b) How many points should we assign a correct answer or a wrong answer so that the expected score is 0 for a student who guesses all the answers.
6. Show that $E[X + Y] = E[X] + E[Y]$ for any two random variables X and Y .
7. If X and Y are independent, then show that $E[X \times Y] = E[X]E[Y]$.
8. If X and Y are independent, then show that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Exercise sheet 6

Probability and Statistics, MTH102

1. For a random variable X which only takes positive values, show that $aP\{X \geq a\} \leq E[X]$ (*Hint*: why is $E[X] = \sum x_i p(x_i) \geq \sum_{i \text{ so that } x_i \geq a} ap(x_i)$?)

How do you use that?)

2. Note that if $|X - E[X]| \geq k$ then $(X - E[X])^2 \geq k^2$.

(a) Use this observation and the previous exercise to show that $P\{|X - E[X]| \geq k\} \leq \frac{\text{Var}(X)}{k^2}$

(b) Show that if the variance is 0, then the probability that X is exactly the expectation is 1. (*Hint*: Show that the probability that the $|X - E[X]|$ is bigger than any non-zero number (however small!) is 0.)

3. Let $X_1, X_2, \dots, X_n$ denote independent and identical distributions. Since they are identical, $E[X_1] = E[X_2] = \dots = E[X_n]$, and let us denote this common expectation by μ . Similarly, $\text{Var}(X_1) = \text{Var}(X_2) = \dots = \text{Var}(X_n)$, and we denote the common variance by σ^2 . Use the above to show that $P\{|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu| \geq k\} \leq \frac{\sigma^2}{k^2 n}$

4. Someone claims that a particular coin is biased toward getting a head with probability 0.7. Suppose you allow an error of at most 0.01, apply the previous exercise to estimate the number of times you should toss the coin to be 95% sure that the claim is true. (*Hint*: Assuming that the probability of getting a head is 0.7, what should the expected value be? If you performed the experiment n times, and the average number of heads differed from 0.7 by at most 0.01, what does the previous question tell you about its probability?)

Exercise sheet 7

Probability and Statistics, MTH102

1. The average number of typos in a certain newspaper is 0.3. What is the probability that the newspaper has more than 3 typos?
2. The probability of winning in a particular game is 0.01. Use the Poisson approximation to compute the probability of winning in 5 out of 10000 games.
3. If X is a Poisson random variable with parameter λ , then show the following relations on the moments: $E[X^n] = \lambda E[(X+1)^{n-1}]$.
4. If X is a Poisson random variable with parameter λ , then $p(i)$ increases and then decreases. Compute $p(i+1)/p(i)$ and use that to find out for which value of i the $p(i)$ will attain its maximum value.
5. If X is a continuous random variable with density function $f_X(x)$ defined so that $f_X(x) = c$ as long as $a < x < b$, and 0 otherwise, then what should the value of c be so that it is a valid density function?
6. If X and Y are continuous random variables, prove the following counterparts of discrete random variables:
 - (a) $E[aX + b] = aE[X] + b$ for any numbers a and b
 - (b) $E[X + Y] = E[X] + E[Y]$
 - (c) $\text{Var}(aX + b) = a^2\text{Var}(X)$
7. Let X be a *continuous* random variable that cannot take negative values. Show the Markov's inequality, i.e. $P\{X \geq k\} \leq \frac{E[X]}{k}$. Why does the Weak Law of large numbers hold for continuous random variables too?

Exercise sheet 8

Probability and Statistics, MTH102

1. Let X denote a normal random variable with mean μ and variance σ^2 . Remember that $\Phi(x)$ is the cumulative function for the *standard* normal distribution. Prove that $F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$
2. Let X denote a normal random variable with mean μ and variance σ^2 . Show that
 - (a) $P\{X > \mu + \sigma\} = 1 - \Phi(1)$
 - (b) $P\{\mu < X < \mu + \sigma\} = \Phi(1) - \Phi(0)$
 - (c) $P\{\mu - \sigma < X < \mu\} = \Phi(0) - \Phi(-1)$
 - (d) $P\{\mu - 2\sigma < X < \mu - \sigma\} = \Phi(2) - \Phi(1)$
 - (e) $P\{X < \mu - 2\sigma\} = \Phi(-2)$
 - (f) In general, derive a formula for $P\{\mu - a\sigma \leq X \leq \mu + b\sigma\}$ in terms of Φ and observe that it does not depend on the parameters μ or σ .
3. If the average total marks of a particular class is 50%, and the standard deviation is 20%, then assuming that it is modelled by a normal distribution, estimate the percentage of students who will get an A grade if the cutoff for an A is 95%.
4. Show that if the cutoff for an A is set at the mean plus standard deviation for any course, then assuming that it is modelled by a normal distribution, the percentage of students earning an A will always be around $1 - \Phi(1)$
5. Consider a random variable with probability density function $f_X(x) = \lambda e^{-\lambda x}$ when $x \geq 0$ and 0 when x is negative.
 - (a) Show that f_X is a valid density function
 - (b) Find $E[X]$ and $\text{Var}(X)$
 - (c) Show that X satisfies the property that $P\{X > a + b | X > a\} = P\{X > b\}$. That is that assuming that we know that X is bigger than a , then the probability that X is bigger than $a + b$ is the same as the probability that X is bigger than b .
6. Let X denote the uniform distribution on the interval $(0, 1)$. Let Y denote the random variable satisfying $Y = X^3$. Find a probability density function for Y . (*Hint:* You will need to use the cumulative function and its interpretation as a probability)

7. It is known that for very large n the binomial distribution is approximated by the normal distribution with the same mean and variance (Unlike the Poisson, we do not need p to be relatively small, and also unlike Poisson this time it is continuous).
- (a) Use this to find the probability of getting more than 90% heads if the probability of getting a head is 0.7 in n tosses where n is very large. Leave your answer in terms of Φ .
 - (b) In general, if X is binomial random variable with n very large, what do you think the following probability should approximately be in terms of Φ ?

$$P \left\{ a < \frac{X - np}{\sqrt{np(1-p)}} < b \right\}$$

Exercise sheet 9

Probability and Statistics, MTH102

1. Let $f(x, y)$ be a joint density function for the random variables X and Y . Express $P\{X > a, Y < b\}$ and $P\{X > a, Y > b\}$ in terms of the density function and the marginal density functions.
2. Consider a uniform random variable defined on an interval (a, b) . For what values of c and d will the random variable $Y = cX + d$ be a uniform random variable on $(0, 1)$?
3. Let X and Y denote the uniform random variables, uniform on the interval $(0, 1)$. Compute the probability density function of the sum $X + Y$.
4. If two people plan to meet and their probability of arriving between say 9am and 10am is uniformly and independently distributed, compute the probability that one of them will have to wait for at least 5 minutes.
5. Show that the sum of two binomial distributions with parameters (m, p) and (n, p) is also a binomial distribution with parameters $(m + n, p)$. Use the following identity from combinatorics:

$${}^{m+n}C_k = \sum_{i=0}^n {}^nC_i {}^mC_{k-i}$$

Exercise sheet 10

Probability and Statistics, MTH102

1. Compute the moment generating functions of the Poisson distribution, binomial distribution, and the normal distribution.
2. Use the moment generating functions to show that the sum of two Poisson distributions is also a Poisson distribution.
3. If you assume a coin is fair, and you toss it 1000 times, use the central limit theorem to estimate the probability that you get more than 70% heads. Compare it with the answer that you will get if you were to use the the law of large numbers.
4. Show that following properties of $\text{Cov}(X, Y)$
 - (a) $\text{Cov}(X, X) = \text{Var}(X)$
 - (b) $\text{Cov}(X, Y) = \text{Cov}(X, Y)$
 - (c) $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$
 - (d) $\text{Cov}(X_1 + Y_2, Y_1 + Y_2) = \text{Cov}(X_1, Y_1) + \text{Cov}(X_1, Y_2) + \text{Cov}(X_2, Y_1) + \text{Cov}(X_2, Y_2)$
 - (e) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.